Reconstruction-Enhanced Prototype Network for Hyperspectral Image Open-Set Classification

Ruijie Xu, Jiayu Yu, Xianghai Cao, *Member, IEEE*, Jing Zhao, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—With the gradual maturity of deep learning technology and its extensive application in the field of remote sensing, hyperspectral image (HSI) classification technology has made tremendous progress. The existing methods can achieve excellent classification performance in a closed-set environment (CSE), where the class distribution of the training and test set is consistent. However, in the open environment of the real world, many samples in the test set belong to the classes that never appear in the training set. Therefore, when unknown samples are present, existing closed-set classification methods will predict them as one of the known classes and cannot identify the unknown samples. Open-set recognition (OSR) aims to solve this problem by simultaneously identifying unknown classes and distinguishing known classes. In this article, we propose a reconstruction-enhanced prototype (RePro) network for HSI open-set classification. Specifically, the prototypes are constructed for each class in the embedding space, and the classification loss is also introduced to make the samples closer to the prototype of each class. Then the reconstruction loss based on L_1 is adopted to fully exploit the spectral–spatial information of training samples to provide the auxiliary information to identify the samples of unknown classes. Finally, the reconstruction loss and prototype distance are combined to identify samples of unknown classes. Experiments on multiple datasets demonstrate the effectiveness of the proposed method.**

***Index Terms*—Hyperspectral imagery, open-set, prototype, reconstruction.**

I. Introduction

HYPERSPECTRAL remote sensing, as a new type of information collection method that combines imaging and spectral technologies, has become an increasingly popular technique for remote sensing and Earth observation applications due to its rich spectral and spatial information [1]. Hyperspectral image (HSI) classification can be applied to many fields, such as precision agriculture, military applications, environmental monitoring, and food safety inspection.

In the early days, manually extracted spectral or spectral–spatial features were combined with classical classifiers, such as support vector machine or random forest to obtain classification results [2], [3]. However, intraclass spectral variability caused by illumination variations, weathering, and material composition differences, often leads to significant spectral variations within the same land cover class.

With the rapid development of deep neural networks, HSI classification technology has made great progress. Among them, the convolutional neural network (CNN), due to the powerful feature extraction capability, constantly refresh the classification accuracy [4], [5], [6], [7]. He et al. [8] proposed the residual network (ResNet), which uses a shortcut connection that brings excellent feature extraction capabilities while alleviating overfitting. It has also achieved outstanding results in the hyperspectral domain [9], [10]. However, due to the limitations of its local receptive field, the transformer architecture based on the global self-attention mechanism has emerged [11], [12]. Zhong et al. [13] proposed to replace the ResNet with transformer in SSRN. Sun et al. [14] proposed a transformer for spectral–spatial feature learning, and a series of transformer-based methods have achieved excellent performance in the field of HSI classification [15], [16], [17]. Besides, Li et al. [18] proposed an end-to-end deep learning framework that jointly optimizes both the denoising and classification tasks through a compound loss function. Zhe et al. [19] proposed a combined classification method for HSI, where the t-distributed stochastic neighbor embedding (t-SNE) dimensionality reduction algorithm is integrated with a deep learning network to achieve accurate classification of ground cover and roof materials. Li et al. [20] proposed an HSI classification model based on the Mamba model, which can integrate spatial and spectral information adaptively while simultaneously model long-range interaction across the entire image. In cross-scene HSI classification, Zhang et al. [21] proposed a single-source domain extension network (SDEnet) to enhance the reliability and effectiveness of domain extension. Zhao et al. [22] proposed an unsupervised domain adaptation (UDA) framework, which can align task-related features and learn task-specific decision boundaries in parallel. Zhang et al. [23] proposed SpectralX, which adapts existing RSFMs as backbones and adopts a two-stage training strategy to effectively handle diverse spectral inputs, achieving improved domain generalization. Meanwhile, they also proposed a bidirectional domain adaptation (BiDA) framework for cross-domain HSI classification, which enhances both adaptability and class separability in the target domain [24].

Some unsupervised learning methods are also used for HSI classification. Mei et al. [25] proposed an unsupervised spatial–spectral feature learning method using 3-D convolutional autoencoder for HSI classification. Mirsharji and Ghassemian [26] combined spatial feature extraction with spectral feature extraction and achieved superior classification performance using masked autoencoders.

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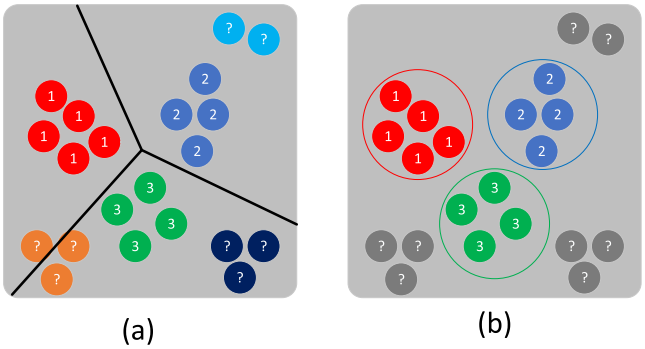


Fig. 1. (a) Closed-set methods classify the unknown samples into one of the known classes. (b) Open-set methods can correctly classify known classes and identify unknown samples.

Zhu et al. [27] proposed a graph-based nonlocal total variation method for unsupervised classification of HSIs. Nalepa et al. [28] combined 3-D convolutional autoencoders with clustering to segment HSIs in a fully unsupervised way. Hassanzadeh et al. [29] proposed a multimanifold spectral clustering algorithm based on contractive autoencoders for the unsupervised classification of hyperspectral imagery.

Although these methods can achieve high classification accuracy, they are all based on the closed-set assumption, meaning that all training and test data share the same label space. However, in the real world, due to the diversity of land cover types, the test set often includes unknown classes that do not appeared in the training set. In this case, unknown samples will be misclassified as one of the known classes, as shown in Fig. 1(a).

To solve this problem, open-set classification is proposed, where the unknown classes can be identified and the known classes can be classified correctly simultaneously. OpenMax [30] is the first method to solve this problem in the context of deep learning, which enables the network to estimate the probability that an instance belongs to unknown class. It uses the activation vector (the last output feature before SoftMax) to estimate the distance distribution of each class and introduces an additional class that includes all the samples of unknown class. Scheirer et al. [31] used a "1-vs-set machine" model to classify known samples correctly while rejecting unknown samples. Wang et al. [32] used attention mechanism to learn different features of samples for open-set classification. Kong and Ramanan [33] used a trained generator to synthesize a large number of fake unknown samples, increasing the presence of outliers during training to improve the prediction of unknown samples. Neal et al. [34] proposed using counterfactual images generated by generative adversarial networks as unknown samples to train the classifier. Pal et al. [35] proposed an open-set classification method of generating adversarial network based on meta-learning, which can be applied in the hyperspectral field to distinguish open space from closed space. Sun et al. [36] proposed a spectral–spatial multiple layer perceptron (SSMLP) network with reciprocal points learning to improve the classification performance. Xu et al. [37] used generative adversarial networks and mutual-point learning to address the challenges of a limited number of classes and low openness.

Nowadays, some methods based on prototype network have become popular in the field of open-set classification. Yang et al. [38] proposed a convolution prototype network to solve the open-set classification problem by matching the features extracted by CNN with the prototype of each known class. Liu et al. [39] learned multiple Gaussian prototypes for each class to represent the data distribution in latent space. Huang et al. [40] integrated the feature learning ability of autoencoders with prototype networks, achieving good results in open-set classification. Pal et al. [41] proposed the residual 3-D convolution block attention module to extract the discriminative prototypes for each known class, and then designed an outlier calibration network based on meta-learning to distinguish unknown samples. Xie et al. [42] proposed a prototype network based on feature consistency, which constructs a scalable prototype set using the extracted features and identifies known and unknown samples according to a prototype guided open-set module (POSM). Chen et al. [43] proposed a fractional Fourier-based frequency–spatial–spectral prototype network for open-set classification of HSI. Although these methods have achieved good results in the open-set classification of traditional images, but in the hyperspectral field, it is difficult to determine an accurate prototype point for each class due to the limited number of labeled samples, and it cannot fully use the rich spectral–spatial information of training data.

Apart from the prototype network, some reconstruction-based methods have achieved strong performance in open-set classification. Yoshihashi et al. [44] were the first to introduce reconstruction method into open-set classification, which used the potential representation of training samples for reconstruction, achieving robust recognition of unknown classes without compromising the accuracy of known classes. Liu et al. [45] used a multitask deep learning method to simultaneously classify and reconstruct HSI, in which the classification task estimates the probability of known classes, and the reconstruction task evaluates the score for unknown classes. Baghbaderani et al. [46] first extracted features from the samples, and then performed reconstruction in the embedding space. Yue et al. [47] proposed a model that simultaneously reconstructs spectral and spatial features, which preserves the spectral–spatial information crucial for distinguishing between known and unknown classes. Mu et al. [48] initialized the model using a self-supervised reconstruction task and optimized both the classification and reconstruction tasks with the labeled samples. In addition, they introduced an adaptive threshold module to enhance the recognition capability of unknown samples. The methods based on reconstruction have achieved excellent classification results in the hyperspectral field, because it can make full use of the spectral–spatial information of hyperspectral data. However, for some samples that are highly similar to known classes but belong to unknown classes, there is a tendency to misclassify them as known, which reduces the classification accuracy of unknown classes.

Due to the limited labeled samples and intraclass spectral variation, hyperspectral open-set classification is

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a challenging task. The prototype network is a reasonable choice for HSI open-set classification, as the prototype represents the center of each class in the embedding space. However, with a limited number of labeled samples, the prototype cannot accurately represent the distribution of each class. Methods based on reconstruction can fully exploit the spectral–spatial information of training samples; however, due to the large intraclass spectral variation, they also struggle to identify the unknown samples easily. So, we propose a reconstruction-enhanced prototype (RePro) network in this article to combine the merits of these two methods.

The main contributions of our work can be summarized as follows.

1) We propose a novel method named RePro for HSI classification with unknown classes. The proposed method can identify unknown classes and classify the known class with improved accuracy.
2) For the training of prototype network, the classification loss is also introduced to make known samples closer to the prototype of each class.
3) The reconstruction loss and the class prototype are combined to improve the classification of unknown samples. The proposed model combines the merits of both the prototype network and reconstruction loss.

II. Proposed Method

A. Preparation and Problem Statement

We denote D_{tr} as a training set consisting of N labeled instances $\{(x_j, y_j)\}_{j=1}^{N}$, where $x_j \in \mathbb{R}^{H\times W\times M}$ is a hyperspectral patch and $y_j \in \{1, \ldots, K\}$ is its class label. For the test set, it includes the known class D_k and the unknown data D_u, with a label of $(K+1)$. The aim of open-set recognition (OSR) is to achieve the following goals based on D_{tr}:

$$\begin{cases} \text{if } x_j \in D_k, & \text{then } y_j \in \{1, \ldots, K\} \\ \text{if } x_j \in D_u, & \text{then } y_j = K+1. \end{cases} \tag{1}$$

B. Overall Framework

In this article, we propose a new open-set classification method, which consists of three modules: HSI prototype module, HSI reconstruction module, and open-set recognition module. First, the prototype module uses the embedding features to learn the prototype points for each class. To further increase the compactness of each class in the embedding space, the classification loss is also introduced to optimize the network parameters. Then, the spectral reconstruction module is used to fully exploit the spectral–spatial information of the training set. After training, the distance between test samples and prototypes, and the reconstruction error are combined to detect unknown samples. The class label can be predicted using the trained classifier. The overall framework of the proposed method is shown in Fig. 2.

C. HSI Prototype Module

In this article, a prototype refers to feature vector of a class, representing the class distribution in the feature space. Classification is performed based on the distance between a sample and its prototype vector.

The feature extraction network can be represented by the function $F : X \to Z$. To extract the features of HSI samples and build accurate prototype points, the feature extraction network consists of three encoder modules, each of which contains two residual blocks and an average pooling layer. The residual blocks alleviate the gradient vanishing problem by introducing skip connections, improving the performance of deep neural networks. Each residual block contains two convolutional layers, followed by a batch normalization layer and a ReLU activation function. We take $x \in X$ as input, where X represents the training sample space. After feature extraction, we obtain its embedding features

$$z = F(x). \tag{2}$$

During the training process, we calculate the corresponding prototype points using the training samples of each class, as shown in (3), where m_i represents the prototype point of class i, and x_{ij} represents the jth sample of class i

$$\boldsymbol{m}_i = \frac{1}{N_i}\sum_{j=1}^{N_i}\left(F\left(x_{ij}\right)\right) \quad i \in \{1, \ldots, K\}, j \in \{1, \ldots, N_i\}. \tag{3}$$

Then we optimize the model's parameters by minimizing the distance between the latent representations of the samples and their corresponding prototype points, as shown in (4), where K represents the number of known classes, $F(x_{ij})$ represents the embedding feature of each sample, and N_i and m_i represent the number of samples and the prototype point of the ith class, respectively. The optimized model parameters ensure that the latent representations of each known sample are tightly clustered around its prototype point. Through this approach, we can obtain prototype points in the feature space that can effectively distinguish known and unknown classes

$$L_p = \sum_{i=1}^{K}\sum_{j=1}^{N_i}\left|\left|F\left(x_{ij}\right) - \boldsymbol{m_i}\right|\right|. \tag{4}$$

In addition, to further increase the compactness of each class in the embedding space, a classifier with cross-entropy loss is also introduced to optimize the model parameters. After feature extraction, the feature dimension of a sample is (1, 1, 512), and then its features are mapped through a fully connected layer (1, 1, classes), and finally, the softmax function is used for classification. The classification loss L_c is as follows, where y' represents the value of the real label after one-hot encoding, and $\tilde{y}$ is the label predicted by softmax

$$L_c\left(y', \tilde{y}\right) = -\sum_{j=1}^{N} y_j' \log\left(\tilde{y_j}\right). \tag{5}$$

In this prototype module, we weigh the prototype loss and classification loss to obtain the total loss, and then minimize the total loss by backpropagation, so that the model can accurately obtain the prototype points of each class while correctly classifying samples, where β is taken as 0.5

$$\text{Loss} = \beta L_p + (1-\beta) L_c. \tag{6}$$

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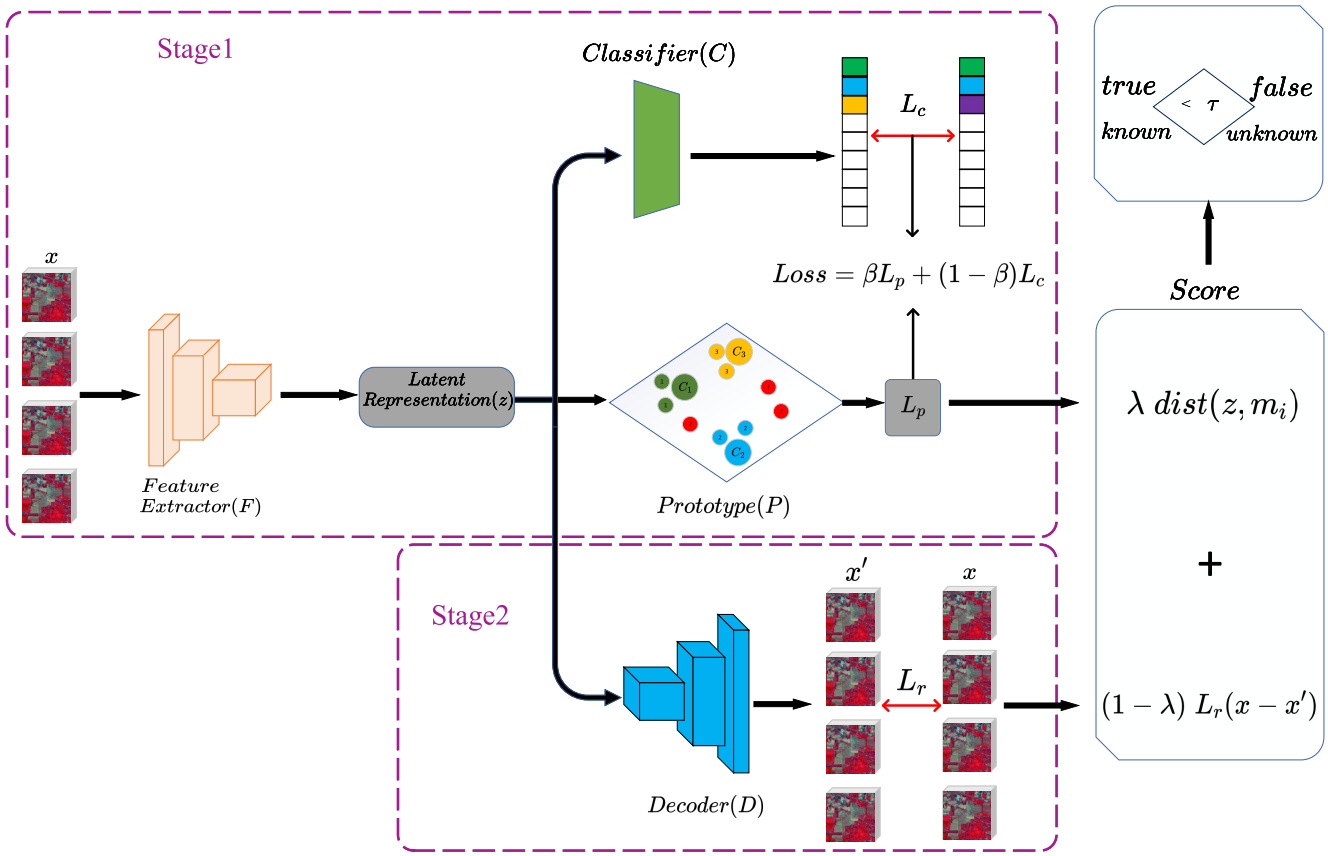


Fig. 2. Flowchart of the proposed method. (Left) Training phase of the proposed method consists of two parts: prototype classification and reconstruction modules. (Right) Testing phase of the proposed method, combining prototype distance and reconstruction error to determine the class of input samples.

D. HSI Reconstruction Module

In the HSI reconstruction module, we fix the parameters of the feature extraction module $F(\cdot)$ and only optimize the decoder parameter $D(\cdot)$. The HSI reconstruction module attempts to recover the original input x from the latent feature z. Where $\tilde{x}$ is the reconstructed instance, $D(\cdot)$ is the reconstruction function, and z is the embedding feature extracted through $F(\cdot)$

$$\tilde{x} = D(z). \tag{7}$$

Here, we use L_1 distance as the reconstruction loss. By optimizing the reconstruction loss, we can make full use of the spectral–spatial information of training samples

$$L_r(x, \tilde{x}) = \sum_{j=1}^{N} \left|\left|x_j - \tilde{x_j}\right|\right|_1. \tag{8}$$

The reconstruction network consists of three decoder modules, each of which includes two deconvolution layers. Each deconvolution layer is followed by a batch norm and ReLU activation function, and finally, the samples are restored to their original size.

E. Open-Set Recognition

After the training phase, we obtain the optimized parameters of $F(\cdot)$ and $D(\cdot)$. For a test sample x, which can belong to either the known or unknown classes, its embedding feature z can be obtained by the feature extraction module $F(\cdot)$. We first calculate the minimum distance between the sample and all the prototype points

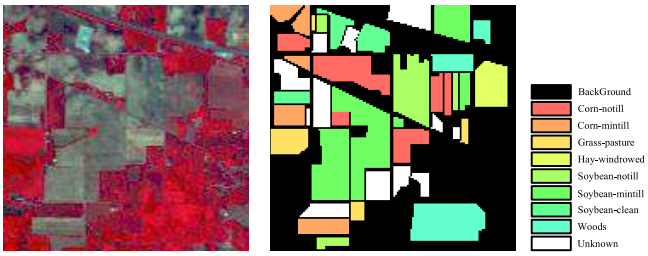
$$\text{dist} = \min_{i=1}^{K} \|z - \boldsymbol{m}_i\|. \tag{9}$$

Then, the reconstruction loss L_r is calculated by the HSI reconstruction module, and the score is obtained by weighted average of the prototype distance and reconstruction loss

$$\text{Score} = \lambda \text{dist} + (1-\lambda) L_r. \tag{10}$$

Finally, if the score is greater than the threshold τ, the test sample is identified as unknown class; otherwise, it means that the sample is a known class, and the real class is obtained through the classification module. Where τ is decided by ensuring 85% of data in the test set to be recognized as known, following the setting adopted in several recent open-set recognition studies [32], [39]. This configuration is particularly suitable for HSI data, where the high spectral similarity among classes increases the likelihood of confusion. Setting τ to achieve an 85% known-class recognition rate provides a conservative balance—prioritizing the accuracy of

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(a) (b)

Fig. 3. Indian pines dataset. (a) False-color map. (b) Ground truth.

TABLE I
SAMPLE DISTRIBUTION OF INDIAN PINES

Index	Name	Number of Samples
1	Corn-notill	1428
2	Corn-mintill	830
3	Grass-pasture	483
4	Hay-windrowed	478
5	Soybean-notill	972
6	Soybean-mintill	2455
7	Soybean-clean	593
8	Woods	1265
9	Unknown	1745
	Total	10249

known-class classification while preserving the ability to detect meaningful unknown samples

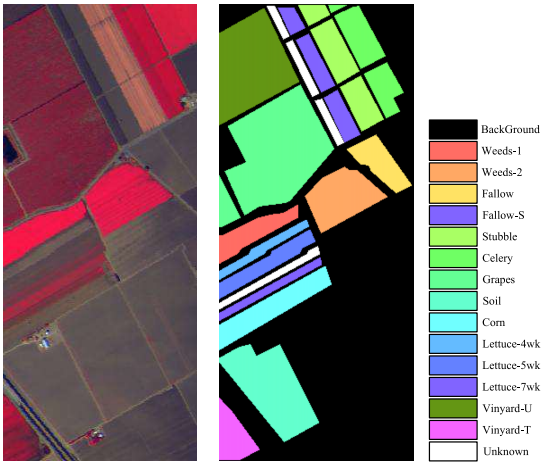
$$y_{\text{pred}} = \begin{cases} \text{Unknown}, & \text{socre} > \tau \\ \text{Label of } x, & \text{otherwise.} \end{cases} \tag{11}$$

III. EXPERIMENTS

A. Datasets

1) Indian Pines: Indian Pines was taken by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in northwestern Indiana. The image has a size of 145 × 145 and consists of 220 bands of which 20 bands subjected to water vapor absorption were discarded and eventually only 200 bands were used. This dataset has annotated a total of 16 classes, most of which are agricultural land cover. Due to the small sample size of some classes, we treated them as unknown classes and ultimately chose eight classes as known, as shown in Table I. Among them, the unknown classes include Alfalfa, Corn, Grass-trees, Grass-pasture-mowed, Oats, Wheat, Buildings-Grass-Trees-Drives, and Stone-Steel-Towers. The false-color and ground-truth maps of this dataset are shown in Fig. 3.

2) Salinas: Salinas was also taken by the AVIRIS sensor, which is an image of the Salinas Valley in California, USA. The image size is 512 × 217, with a total of 224 bands. However, we use the remaining 204 bands after removing the 108–112th, 154–167th, and the 224th band that are absorbed by water. The ground truth consists of 16 classes, including fallow land, celery, etc. We select Fallow-rough-plow and Lettuce-6wk as the unknown classes, and the others as known



(a) (b)

Fig. 4. Salinas dataset. (a) False-color map. (b) Ground truth.

TABLE II
SAMPLE DISTRIBUTION OF SALINAS

Index	Name	Number of Samples
1	Weeds-1	2009
2	Weeds-2	3726
3	Fallow	1976
4	Fallow-smooth	2678
5	Stubble	3959
6	Celery	3579
7	Grapes	11271
8	Soil	6203
9	Corn	3278
10	Lettuce-4wk	1068
11	Lettuce-5wk	1927
12	Lettuce-7wk	1070
13	Vinyard-U	7268
14	Vinyard-T	1807
15	Unknown	2310
	Total	54129

classes, which are shown in Table II, and the false-color and ground-truth maps of the Salinas are shown in Fig. 4.

3) WHU-Hi-HanChuan: WHU-Hi-HanChuan was collected using the Headwall Nano-Hyperspec imaging sensor in Hanchuan City, Hubei Province, China. This dataset covered 1027 × 303 samples with 274 bands in the wavelength range of 400–1000 nm and has a spatial resolution of about 0.109 m. There are a total of 16 classes in this dataset. Due to its large amount of samples, we select Plastic and Bright-object as unknown classes, which are shown in Table III, and the false-color and ground-truth maps are shown in Fig. 5.

B. Implementation Details

To implement the proposed method, we perform it on a computer with Ubuntu 18.04 operating system, equipped

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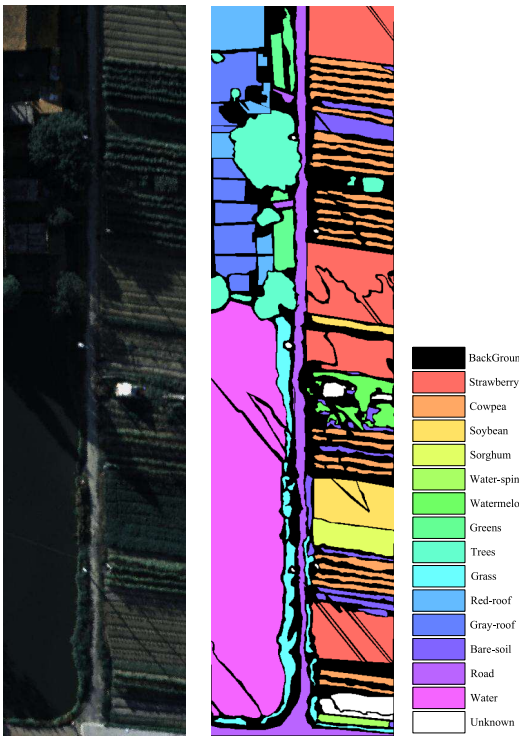


Fig. 5. WHU-Hi-HanChuan dataset. (a) False-color map. (b) Ground truth.

TABLE III
SAMPLE DISTRIBUTION OF WHU-HI-HANCHUAN

Index	Name	Number of Samples
1	Strawberry	44735
2	Cowpea	22753
3	Soybean	10287
4	Sorghum	5353
5	Water-spinach	1200
6	Watermelon	4533
7	Greens	5903
8	Trees	17978
9	Grass	9469
10	Red-roof	10516
11	Gray-roof	16911
12	Bare-soil	9116
13	Road	18560
14	Water	75401
15	Unknown	4815
	Total	257530

with two 12-core Intel[1] Xeon[1] Silver 4214 CPU at 2.20-GHz processors, 128 GB of memory, and two NVIDIA GeForce RTX 3090 graphics cards with 24 GB of memory.

In the experiment, for each class, we randomly select 20 samples for training, with the remaining samples being used for testing. We use the same random seeds, ensuring that all the methods use the same training samples. The patch size of the input sample is 16×16, as HSIs contain a large amount of spectral information, and there is often a high degree of correlation between these pieces of information, leading to data redundancy. Directly processing these high-dimensional data not only requires a large amount of computation but also can easily lead to the curse of dimensionality [49]. PCA compression can reduce the dimensionality of the data while preserving the most important information of the data. By selecting the first few principal components, it is possible to preserve the important information of the original data, while ignoring some noise or less important information. The final size of each sample is $16 \times 16 \times 12$. The training consists of 300 epochs with a batch size of 64, and the Adam optimizer is used. The weights for the prototype distance and reconstruction loss are 0.35 and 0.65, respectively, meaning that $\lambda = 0.35$ in (10).

To evaluate performance, we adopt standard metrics for HSI classification: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

C. Results and Analysis

To comprehensively verify the performance of the proposed method on open-set HSI classification, we use both the closed-set and open-set classification methods for comparison.

For closed-set classification, we adopt several classic HSI classification methods, including SVM-RBF, CAE [50], and SSFTT [14]. Among them, SVM-RBF is the most classic classification method. CAE is based on a convolutional autoencoder, while SSFTT tokenizes spectral and spatial features to fully use the capabilities of the transformer to improve classification performance. SVM-RBF cannot predict a novel class because of the closed-set property. For the other methods, we uniformly apply an additional softmax function to their networks and set the softmax threshold to 0.5 to identify the unknown classes.

We also compare our method with specialized open-set classification methods to validate its superiority, including MDL40W [45], SSLR [47], OSR-VIT [51], and SSMLP [36]. Among them, OSR-VIT uses transformer to extract features and then finds a center point for each class, detecting the unknown class by measuring the distance between the sample and the center point. Both MDL40W and SSLR are based on reconstruction for open-set classification tasks. SSMLP uses the spectral space multilayer perceptron network and reciprocal point learning to improve the performance of open-set HSI classification.

For fair comparison, all the comparison methods adopt the optimal settings from the original papers, and all the methods use the same train samples. To ensure the reliability of the experimental results, we repeat ten times for each experiment.

Table IV shows the result of Indian Pines dataset. From the result, closed-set classification methods such as SVM-RBF, CAE, and SSFTT always predict the samples as one of the known classes, leading to poor detection ability for unknown samples. OSR-VIT detects unknown samples by measuring the distance between the sample and the center point, but it cannot fully use the spectral–spatial information of hyperspectral data, resulting in poor classification performance for unknown

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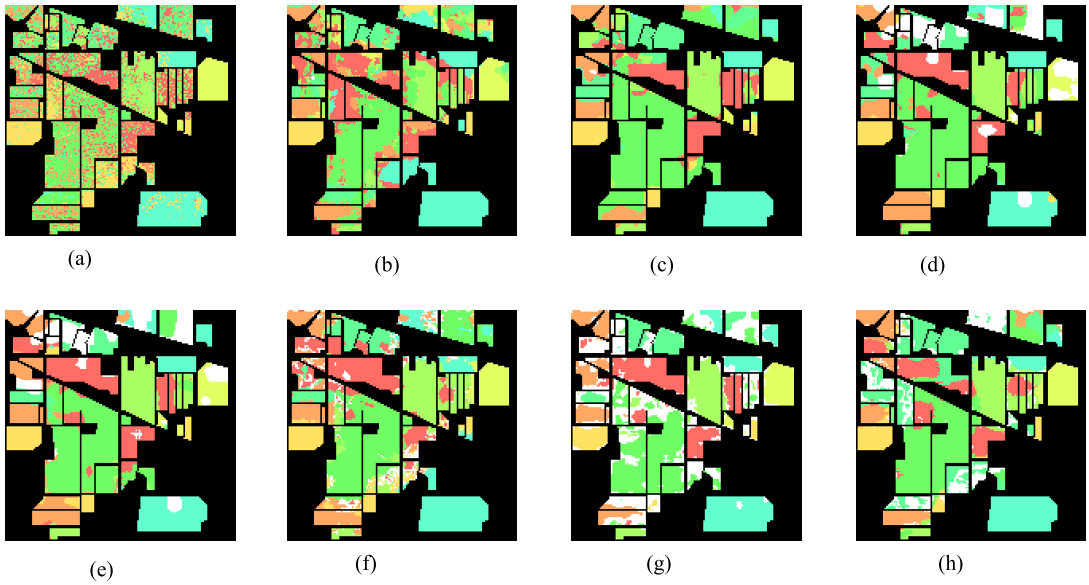


Fig. 6. Classification results of the Indian pines. (a) SVM-RBF. (b) CAE. (c) SSFTT. (d) MDL40W. (e) SSLR. (f) OSR-VIT. (g) SSMLP. (h) RePro.

TABLE IV
CLASSIFICATION PERFORMANCE ON THE INDIAN PINES DATASET

Classes	SVM-RBF	CAE	SSFTT	MDL40W	SSLR	OSR-VIT	SSMLP	RePro
Corn-notill	53.64±1.37	57.49±2.13	68.42±1.85	80.95±2.43	**89.36±0.92**	68.20±1.54	77.52±1.73	65.41±2.36
Corn-mintill	51.33±2.58	74.46±1.87	82.17±1.23	79.04±2.16	**90.00±0.73**	83.13±1.32	70.00±2.85	86.27±1.14
Grass-pasture	88.82±0.83	78.47±2.35	90.48±0.91	95.86±0.63	**96.27±0.53**	92.13±1.07	72.26±3.12	91.93±0.82
Hay-windrowed	99.37±0.32	85.77±2.74	**100.00±0.00**	42.05±3.57	65.69±2.89	**100.00±0.00**	94.35±1.24	99.58±0.41
Soybean-notill	62.45±1.68	57.20±2.25	91.36±0.87	89.81±1.92	**92.39±0.78**	86.11±1.65	79.42±2.13	77.88±1.97
Soybean-mintill	44.85±3.12	63.05±2.47	81.91±1.42	81.55±1.85	81.96±1.63	**82.04±1.38**	59.27±2.94	81.59±1.76
Soybean-clean	55.99±2.87	74.87±1.98	**92.58±0.65**	61.05±3.24	64.59±2.73	78.75±1.87	86.85±1.53	89.88±1.09
Woods	88.85±1.12	90.43±0.95	91.45±0.83	91.78±0.77	**93.52±0.68**	89.33±1.25	93.20±0.72	91.15±0.89
Unknown	0.00±0.00	2.15±0.48	3.56±0.37	24.47±1.84	32.38±1.73	20.11±2.15	50.37±2.62	**53.18±2.34**
OA	51.32±2.15	57.94±1.92	71.14±1.43	71.24±1.87	76.52±1.25	72.07±1.36	71.11±1.58	**77.51±1.12**
AA	60.59±1.83	64.88±1.67	78.02±1.24	71.78±2.03	78.48±1.42	77.76±1.35	75.92±1.63	**81.87±0.97**
Kappa	44.78±2.34	51.58±1.85	66.42±1.57	66.36±1.92	72.60±1.38	67.78±1.43	66.27±1.65	**74.14±1.06**

samples. MDL40W and SSLR are based on reconstruction, but since we treat eight classes as unknown for the Indian Pines dataset, and due to the similarity between some known and unknown samples, it struggles identify unknown classes well through reconstruction alone. The SSMPL method effectively identifies unknown samples by expanding the distance between known and unknown classes using reciprocal points, but it affects its classification performance on known classes. Our method takes into account both prototype distance and reconstruction error and improves the model's ability to recognize unknown classes without affecting the accuracy of closed-set classification. It achieves the best performance in OA, AA, and Kappa. Fig. 6 shows the classification maps for the methods above on the Indian Pines dataset, and it can be observed that our proposed method shows a better classification performance.

Table V shows the classification result of the Salinas dataset. It can be seen that SVM-RBF, CAE, and SSFTT still have a poor detection ability for unknown samples. Compared with MDL40W, SSLR introduces the reconstruction of spatial information, leading to better classification performance than MDL40W. Due to the large number of samples in the Salinas dataset, the OSR-VIT method uses the powerful performance of transformer, outperforming SSLR and MDL40W in classification accuracy for closed-set samples. However, its performance in recognizing unknown samples is inferior to both these methods. Unexpectedly, the SSMLP method achieves optimal results in AA and for unknown class. This may be attributed to its explicit modeling of the open space through reciprocal point learning, as well as the efficient capture of spectral–spatial long-range dependencies by the MLP structure, which enhances its performance. However, its classification performance on the Grapes class is significantly lower than that of our proposed method, resulting in an OA that is still inferior to ours. Fig. 7 shows the classification performance on the Salinas dataset, and it can be observed that the proposed method still achieves the best classification performance.

Table VI lists the classification result for the WHU-Hi-HanChuan dataset. Our method achieves the best scores on

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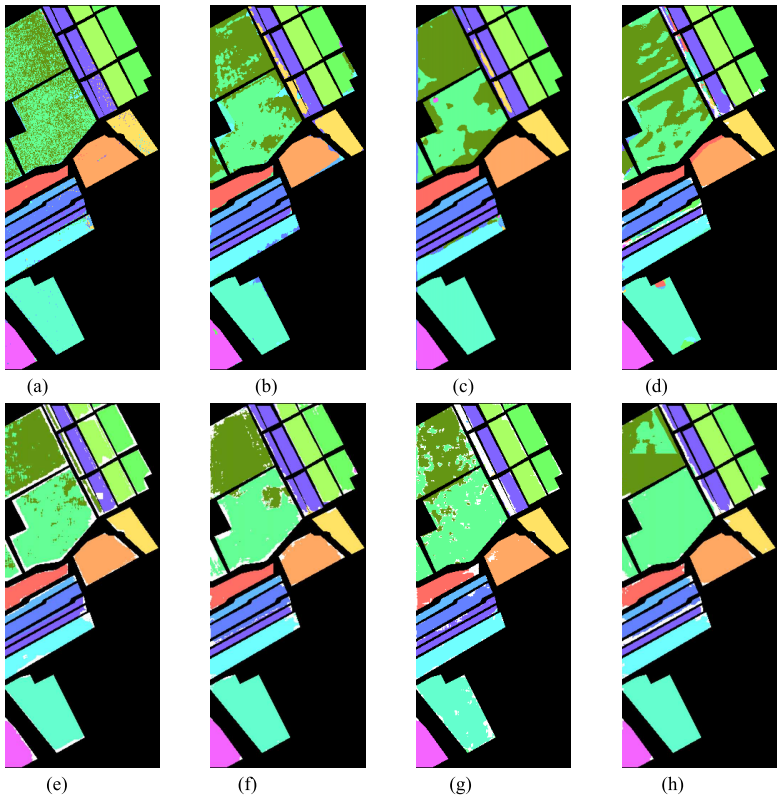


Fig. 7. Classification results of the Salinas. (a) SVM-RBF. (b) CAE. (c) SSFTT. (d) MDL40W. (e) SSLR. (f) OSR-VIT. (g) SSMLP. (h) RePro.

TABLE V
CLASSIFICATION PERFORMANCE ON THE SALINAS DATASET

Classes	SVM-RBF	CAE	SSFTT	MDL40W	SSLR	OSR-VIT	SSMLP	RePro
Weeds-1	99.15±0.23	92.83±1.57	**100.00±0.00**	99.35±0.31	95.87±1.23	98.11±0.67	98.71±0.38	94.67±1.45
Weeds-2	98.68±0.35	99.87±0.12	**100.00±0.00**	90.18±2.13	94.55±1.67	94.82±1.42	91.92±2.05	98.23±0.73
Fallow	87.15±2.37	**100.00±0.00**	99.60±0.18	99.19±0.29	97.12±1.08	98.13±0.65	99.60±0.17	97.93±0.83
Fallow-smooth	95.29±1.23	**100.00±0.00**	98.77±0.54	95.59±1.35	92.08±1.82	98.21±0.63	95.52±1.47	96.56±1.12
Stubble	99.70±0.15	**100.00±0.00**	**100.00±0.00**	99.39±0.21	85.00±2.85	94.11±1.63	91.82±2.13	99.39±0.19
Celery	99.16±0.27	95.05±1.43	**99.22±0.35**	96.84±1.12	86.78±2.37	92.06±1.85	98.38±0.67	96.28±1.24
Grapes	63.76±3.85	73.50±2.97	52.14±4.12	62.08±3.65	79.31±2.35	79.38±2.42	86.05±1.93	**97.62±0.53**
Soil	98.74±0.32	**100.00±0.00**	**100.00±0.00**	93.36±1.87	95.99±1.25	99.10±0.28	98.11±0.63	99.73±0.12
Corn	76.82±2.93	94.57±1.35	93.78±1.42	91.89±1.67	89.08±2.13	94.05±1.37	**97.41±0.85**	94.94±1.28
Lettuce-4wk	92.51±1.53	93.16±1.42	**100.00±0.00**	95.41±1.23	85.21±2.63	95.97±1.15	98.69±0.57	92.42±1.73
Lettuce-5wk	97.66±0.83	**99.95±0.04**	99.79±0.12	97.82±1.03	99.79±0.11	98.34±0.73	98.44±0.68	98.44±0.65
Lettuce-7wk	90.56±1.87	97.57±0.93	**100.00±0.00**	98.04±0.82	95.61±1.42	98.13±0.67	97.94±0.75	96.82±1.13
Vinyard-U	60.97±3.72	63.81±3.45	**95.17±1.12**	82.11±2.83	83.37±2.57	91.00±1.85	82.25±2.63	73.73±3.15
Vinyard-T	98.01±0.73	99.17±0.42	90.65±2.35	**99.72±0.13**	91.64±2.13	98.95±0.53	98.17±0.82	97.45±1.07
Unknown	0.00±0.00	3.62±0.57	3.85±0.43	27.79±2.93	34.07±2.75	15.76±1.87	**64.94±2.36**	50.43±2.85
OA	80.01±1.85	84.32±1.42	84.68±1.37	83.95±1.63	85.86±1.28	88.69±1.15	91.11±0.93	**92.32±0.75**
AA	83.88±1.53	87.54±1.23	88.86±1.07	88.58±1.32	87.03±1.45	89.74±1.18	**93.20±0.82**	92.31±0.97
Kappa	77.78±1.93	82.46±1.57	87.02±1.23	82.19±1.72	84.35±1.43	87.50±1.25	90.12±1.07	**91.42±0.85**

OA, AA, and Kappa. Compared with MDL40W, SSLR, OSR-VIT, and SSMLP, its OA improves by 5.6, 1.69, 8.67, and 8.44 percentage points, respectively. This indicates that both prototype distance and reconstruction error play an important role in open-set classification for HSI. Due to the large number of samples and complex data distribution, the classification performance of MDL40W and SSLR, which rely solely on reconstruction error, is not optimal. Similarly, the classification performance of OSR-VIT and SSMLP is also poor with such a number of samples. However, our model can still effectively detect unknown samples and maintain classification accuracy for the known classes. Fig. 8 shows the classification maps on the WHU-Hi-HanChuan dataset. It can be seen that even with a large number of samples, our method can still achieve the best classification performance.

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TABLE VI
CLASSIFICATION PERFORMANCE ON THE WHU-HI-HANCHUAN DATASET

Classes	SVM-RBF	CAE	SSFTT	MDL40W	SSLR	OSR-VIT	SSMLP	RePro
Strawberry	70.52±2.37	87.79±1.53	72.33±2.85	90.04±1.12	95.01±0.93	87.54±1.42	79.02±2.13	**95.23±0.53**
Cowpea	33.93±4.28	67.70±2.73	85.51±1.67	69.96±3.45	74.83±3.12	78.48±2.05	63.33±3.57	**88.29±1.24**
Soybean	64.70±3.15	77.77±2.13	63.50±3.42	82.98±1.87	**91.81±0.85**	87.08±1.35	84.81±1.63	91.11±1.07
Sorghum	91.61±1.25	92.14±1.17	97.27±0.73	59.57±3.82	**98.88±0.32**	83.22±2.23	96.94±0.67	96.23±0.93
Water-spinach	62.33±3.57	96.58±0.83	90.17±1.42	98.58±0.43	99.17±0.27	**99.83±0.12**	96.75±0.95	84.33±2.35
Watermelon	41.63±4.12	42.73±3.95	58.53±3.27	41.03±4.05	75.92±2.53	60.78±3.15	**83.43±1.87**	56.30±3.62
Greens	85.50±1.73	84.16±2.05	80.47±2.37	95.87±0.93	89.38±1.63	**96.98±0.67**	82.82±2.13	90.39±1.42
Trees	44.36±3.82	31.10±4.15	62.47±3.05	53.14±3.57	68.27±2.85	**79.90±2.13**	**79.90±2.07**	68.54±3.23
Grass	31.70±4.23	60.69±3.35	70.30±2.93	60.29±3.45	52.09±3.17	65.66±2.95	**82.00±1.85**	57.33±3.57
Red-roof	80.45±2.13	83.52±1.95	89.95±1.35	68.12±3.25	90.45±1.17	56.86±3.45	**97.44±0.53**	80.47±2.23
Gray-roof	85.05±1.85	59.46±3.57	81.53±2.15	**95.87±0.63**	94.06±1.03	75.29±2.85	94.14±0.93	83.16±2.05
Bare-soil	43.33±3.95	48.51±3.67	66.89±2.73	67.51±3.15	63.55±1.35	64.21±3.25	61.02±3.42	**84.17±1.87**
Road	65.73±3.25	72.14±2.85	70.76±2.63	68.89±3.82	68.08±2.95	76.29±2.45	84.00±1.95	**90.09±1.13**
Water	89.14±1.42	97.54±0.75	92.85±1.25	**98.98±0.18**	97.46±0.93	82.94±2.13	77.87±2.45	95.88±0.85
Unknown	0.00±0.00	2.84±0.57	4.25±0.43	39.75±3.25	40.52±2.93	18.53±1.85	36.88±2.67	**43.93±3.67**
OA	68.22±2.85	76.79±2.13	78.69±1.95	81.84±2.05	85.75±1.42	78.77±1.85	79.00±1.73	**87.44±1.15**
AA	59.33±3.15	66.98±2.67	72.45±2.35	72.71±2.53	80.23±1.85	74.24±2.13	80.02±1.73	**80.36±1.62**
Kappa	63.42±2.95	72.79±2.25	68.32±2.63	78.96±2.35	83.54±1.57	75.49±2.05	76.11±1.95	**85.40±1.25**

From the experimental results, it is clear that our method does not achieve the best performance on all the known classes. However, compared with other open-set methods, it demonstrates a more balanced classification performance, effectively recognizing unknown samples without significantly compromising the accuracy on known classes. For example, the SSLR achieves relatively high classification accuracy on known classes in the Indian Pines dataset, but its accuracy on unknown classes is only 32.38%, which is significantly lower than that of our method. Moreover, it shows notably poor performance on the Hay-windrowed and Soybean-clean classes. In the Salinas dataset, due to the limited number of unknown classes, closed-set methods such as CAE and SSFTT achieve good classification results on known classes. However, they perform poorly in recognizing unknown samples. In the WHU-Hi-HanChuan dataset, while SSLR, MDL40W, and SSMLP achieve the best classification results for certain known classes, but their performance on some other classes is poor. In contrast, our method achieves more balanced results and outperforms others in terms of overall classification accuracy.

Meanwhile, Fig. 9 further validates the superiority of the proposed method through t-SNE visualization. As shown, the RePro method effectively clusters samples from known categories into compact and clearly separated groups, while distinctly isolating samples of unknown categories from the known ones. In contrast, other methods exhibit notable overlap between known and unknown classes, resulting in indistinct decision boundaries and degraded open-set recognition performance.

D. Parameter Analysis

1) Analysis of λ and β: Prototype distance encourages the model to make samples of known classes as close as possible to their corresponding prototype points, thereby improving the recognition accuracy for known classes. On the other hand, reconstruction error helps the model better reconstruct samples from known classes, but its ability to reconstruct samples from unknown classes is limited. Compared with traditional images, HSIs contain rich spectral and spatial features, which results in different roles for prototype distance and reconstruction error in open-set classification tasks for HSI. Therefore, by setting λ to different values, we can observe the impact of reconstruction and prototype of the RePro model. In addition, in the prototype module's loss function, β represents the weight ratio between prototype distance and classification loss. Different values of β directly affect the model's classification performance.

TABLE VII
PARAMETER ANALYSIS OF λ AND β ON INDIAN PINES

β	λ				
	0	0.35	0.55	0.75	1
0.1	70.31	72.23	71.83	68.24	65.43
0.3	73.52	75.42	74.21	70.84	67.15
0.5	71.44	**77.51**	73.88	69.12	66.97
0.7	75.63	76.44	72.83	68.52	65.24
0.9	74.31	74.28	70.53	66.73	63.92

Tables VII–IX show the performance of the RePro model on the Indian Pines, Salinas, and WHU-Hi HanChuan datasets under different λ and β values. The experimental results indicate that on three datasets, the model achieved the best performance when λ was set to 0.35 and β was set to 0.5.

Specifically, $\lambda = 0.35$ means that the prototype distance accounts for 35% of the weight, while the reconstruction error contributes 65%. This shows that reconstruction error plays a more critical role in HSI open-set classification. When λ is set to 0, the model degenerates into a reconstruction-based

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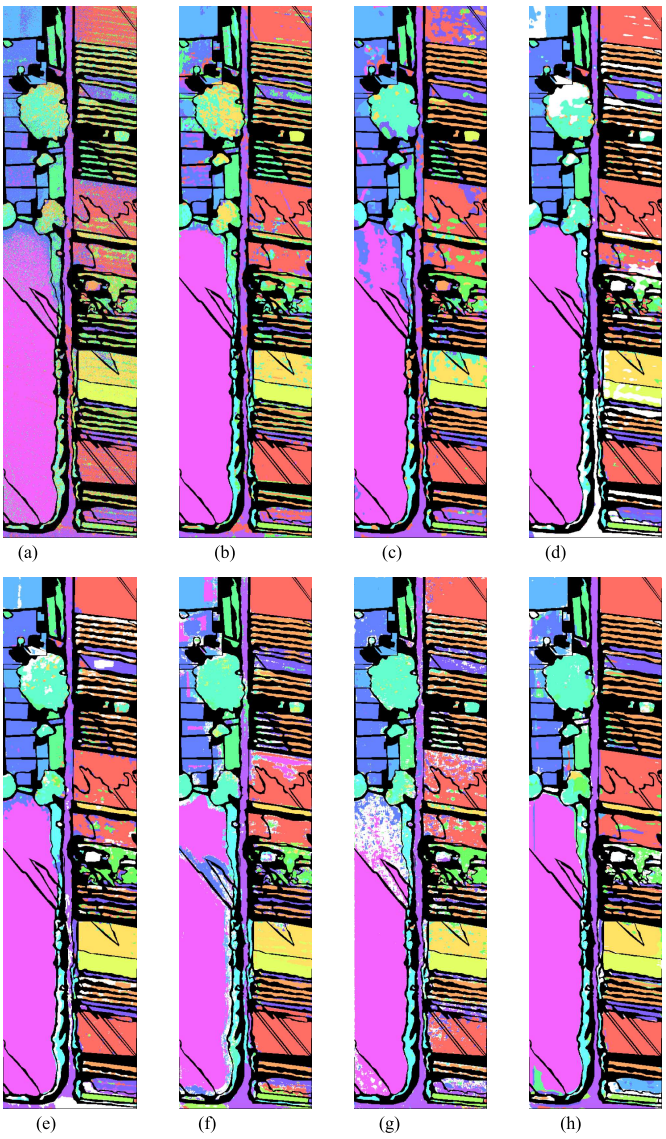


Fig. 8. Classification results of WHU-Hi-HanChuan. (a) SVM-RBF. (b) CAE. (c) SSFTT. (d) MDL40W. (e) SSLR. (f) OSR-VIT. (g) SSMLP. (h) RePro.

TABLE VIII
PARAMETER ANALYSIS OF λ AND β ON SALINAS

β	λ				
	0	0.35	0.55	0.75	1
0.1	86.74	88.84	85.21	82.13	78.91
0.3	89.63	91.08	87.42	84.63	80.34
0.5	84.29	**92.32**	86.96	83.52	79.82
0.7	90.42	90.38	85.74	82.94	78.42
0.9	89.95	89.95	84.32	80.63	77.13

classification method, as the prototype network's enhancement effect is absent, leading to a decline in classification performance. Conversely, when λ is set to 1, the model relies entirely on the prototype network for open-set classification, but the performance declines significantly due to the prototype network's inability to effectively use the spectral–spatial information from the training set.

TABLE IX
PARAMETER ANALYSIS OF λ AND β ON WHU-HI-HANCHUAN

β	λ				
	0	0.35	0.55	0.75	1
0.1	82.41	84.01	83.15	77.53	74.23
0.3	84.23	85.59	85.03	79.82	75.64
0.5	80.58	**87.44**	84.17	78.69	75.48
0.7	84.32	86.21	82.93	77.31	73.84
0.9	83.71	84.34	81.53	75.92	72.43

Setting β to 0.5 indicates that the weights of classification loss and prototype distance are equal during training. This configuration enables the model to consider both prototype distance and classification loss in a balanced manner, thereby achieving stable and high performance across different datasets.

2) Analysis of Patch Size: The spatial patch size S of HSI data cubes has a significant impact on the final performance. Thus, we implement experiments on three HSI datasets to observe the performance varying with the change in patch size S and determine the optimal S. The results are shown in Fig. 10, to balance the performance and computational cost, and we set S to 16.

3) Analysis of the Numbers of Unknown Classes: In the above experiments, the number of unknown classes is smaller than that of known classes. To assess the effectiveness of the proposed method, we conduct experiments on three datasets, selecting different numbers of unknown classes. The classification performance, measured by OA, is compared across different methods, with the results shown in Fig. 11.

As shown in the figure, the OA metric for all the methods decreases significantly as the number of unknown classes increases. For closed-set methods, a larger number of unknown classes means fewer known samples. Since these methods inherently lack the ability to recognize unknown samples, their OA remains consistently low. In contrast, open-set methods can identify unknown samples, but their performance is still impacted when faced with a large number of unknown classes. Nevertheless, our method consistently outperforms others in classification accuracy, demonstrating its robustness and effectiveness, as well as its ability to handle the imbalance between known and unknown class distributions.

4) Analysis of the Numbers of Training Samples: In the experiments, we select 20 samples per class to analyze the classification performance of the model. To evaluate the robustness of the proposed method, we conduct experiments with 10, 20, 30, and 40 samples per class across different datasets. This enables us to assess how the model performs under different sample sizes and determine its generalizability. The detailed experimental result is presented in Fig. 12.

The results clearly show that the proposed RePro method consistently achieves the best classification performance under different conditions. This demonstrates the model's high effectiveness, performing optimally even with a limited number of training samples. Moreover, the method shows strong stability

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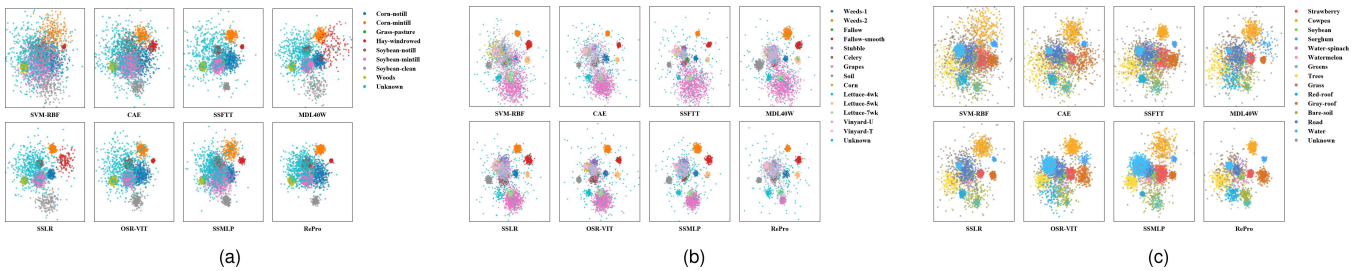


Fig. 9. t-SNE visualization of the open-set recognition results on three datasets. (a) Indian pines. (b) Salinas. (c) WHU-Hi-HanChuan.

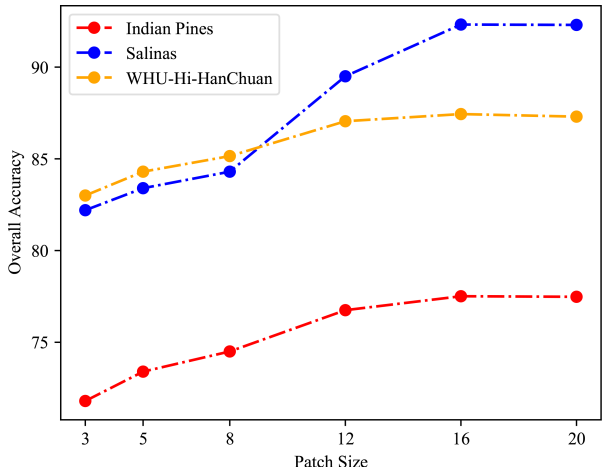


Fig. 10. Influence of patch size S on classification performance.

TABLE X
PARAMETER ANALYSIS OF SPECTRAL DIMENSION C

Metrics		Spectral Dimension C				
		6	12	30	60	original
Indian Pines	OA(%)	75.34	77.51	77.24	76.63	75.98
	Time(s)	658	850	1215	1562	2346
Salinas	OA(%)	89.73	92.32	91.75	90.82	89.49
	Time(s)	1309	1782	2587	3412	4828
WHU-Hi-HanChuan	OA(%)	85.64	87.44	86.72	85.39	84.77
	Time(s)	2546	3128	4362	5323	8242

TABLE XI
ANALYSIS OF COMPUTATIONAL EFFICIENCY

Method	Number Of Parameters	FLOPS	Inference Time
SVM-RBF	1k	1.06k	1.5ms
CAE	24.8k	50k	1.2ms
SSFTT	152k	15.33M	2ms
MDL40W	270k	4.92M	1.2ms
SSLR	300k	46.2M	1.4ms
OSR-VIT	88.1M	17.8G	3.8ms
SSMLP	540k	2.8G	1.8ms
RePro	2.94M	1.02G	2.2ms

and adaptability, ensuring reliable performance regardless of the training sample size.

5) Analysis of the Spectral Dimension: HSIs contain abundant spectral information, but the high correlation between adjacent bands leads to significant data redundancy. This redundancy not only increases the computational complexity of processing high-dimensional data but also exacerbates the "curse of dimensionality" problem. Although deep learning models can automatically extract spectral–spatial joint features through hierarchical learning mechanisms without explicit dimensionality reduction, directly processing the raw high-dimensional data still risks overfitting, especially with limited training samples. Therefore, we conducted experiments with different spectral dimensions to analyze the optimal spectral dimension setting for the RePro model in our experiments.

The experimental result is shown in Table X, which indicate that in the Indian Pines, Salinas, and WHU-Hi-HanChuan datasets, setting the spectral dimension to 12 yields the peak OA, with relatively short training time. In contrast, when the spectral dimension is reduced to 6, although the training time is shortest, the loss of spectral information leads to a decrease in classification accuracy. When the spectral dimension exceeds 12, although classification performance slightly decreases, the model training time increases significantly. Experiment shows that moderate dimensionality reduction effectively removes redundant information, suppresses overfitting, and preserves discriminative spectral features. Thus, we set the spectral dimension to 12.

6) Analysis of Computational Efficiency: To evaluate the practical deployment potential of different methods, we conduct a comprehensive comparison of computational complexity in Table XI, including the number of parameters, FLOPs, and inference time across all the datasets. The proposed RePro framework demonstrates a highly competitive efficiency profile.

SVM-RBF exhibits outstanding efficiency with only 1 K parameters and 1.06 K FLOPs, achieving the fastest inference time of 1.5 ms, making it highly suitable for resource-limited environments. Among deep learning models, CAE and MDL40W maintain a favorable balance between accuracy and efficiency, with the latter achieving 1.2 ms inference despite having 270 K parameters. The transformer-based OSR-ViT shows the highest computational demand (88.1 M parameters, 17.8 G FLOPs) and the slowest inference (3.8 ms), reflecting the heavy cost of attention mechanisms. SSMLP achieves competitive efficiency (540 K parameters, 1.8 ms), while

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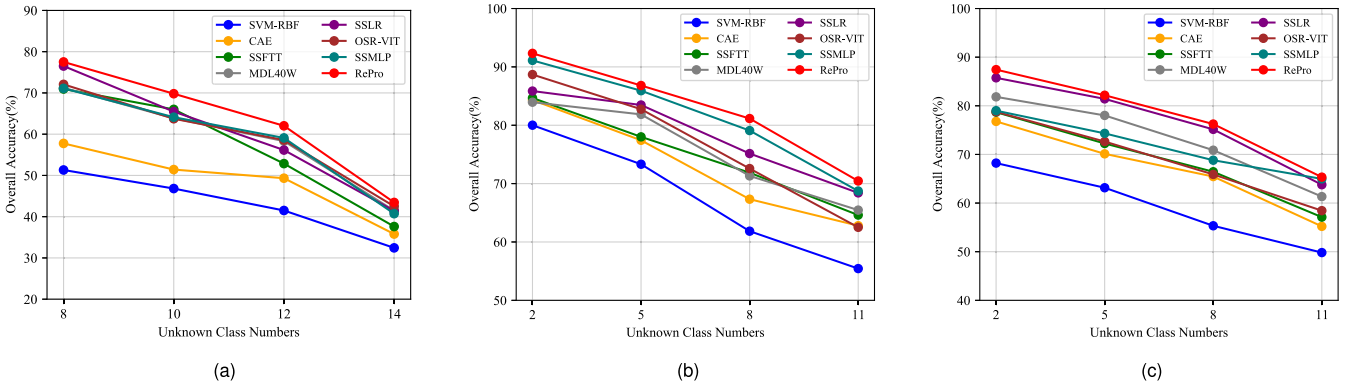


Fig. 11. Classification performance with different numbers of unknown classes. (a) Indian pines. (b) Salinas. (c) WHU-Hi-HanChuan.

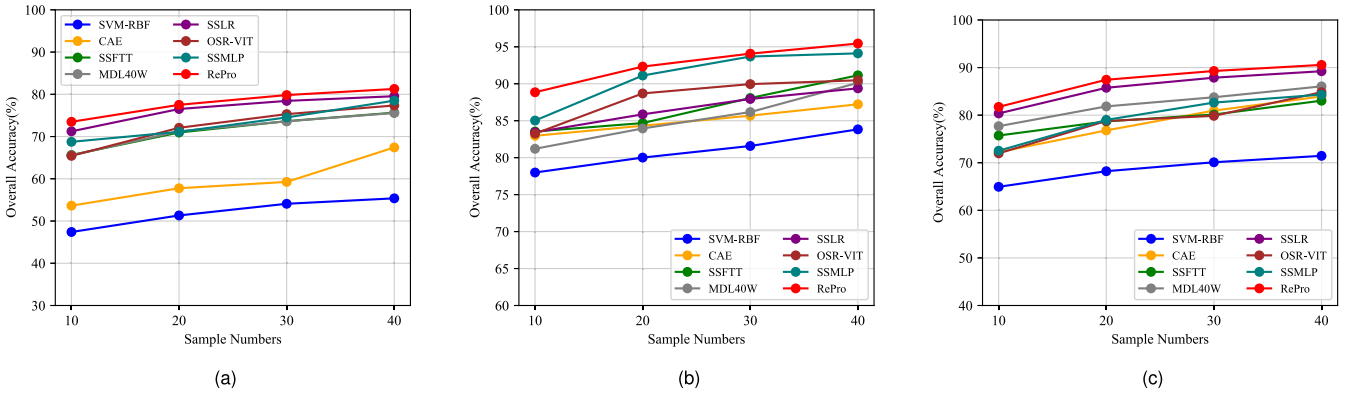


Fig. 12. Analysis of different numbers of training samples. (a) Indian pines. (b) Salinas. (c) WHU-Hi-HanChuan.

RePro attains strong open-set recognition performance with reasonable complexity (2.94 M parameters, 2.2 ms).

IV. Conclusion

Open-set classification is a crucial issue for the HSI classification models to be deployed in practical applications. However, this problem has not received sufficient attention in this field. In this article, we propose a novel open-set classification model—RePro. This model leverages the advantages of both the prototype network and reconstruction. With prototype network, the latent representations of samples from the same class can be gathered together. Through reconstruction, we can fully exploit the spectral–spatial information to identify the unknown samples. The experimental results demonstrate that the proposed method can simultaneously detect unknown samples and classify known ones.

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